Observable Dynamics of Sectorized Observable Frameworks

Deformations, Rate Separation, and Observable Dynamics

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This paper is Part IX of the RIME program. Paper VIII establishes the static object layer of Sectorized Observable Frameworks: SOFs, strict morphisms, categories, and natural accessibility constructions. Paper IX studies observable dynamics: how SOFs deform, how observable trajectories evolve, how different time scales appear, and why different deformation spaces produce different observable dynamics over the same observable architecture.

Abstract

Problem. Paper VIII identifies the Sectorized Observable Framework (SOF) as the common observable architecture in which the RIME observables R_1, R_2, D, and $\mathcal{J}_{\mathrm{acc}}$ arise naturally. The next question is dynamic: how do these observables behave when the SOF itself varies? The difficulty is that different systems deform different data. Rubik deforms generator weights; Yang-like systems deform states; Markov systems deform transition operators; graphs deform adjacency or Laplacian data; quantum systems deform Hamiltonians or gate families.

Approach. We define a deformation of an SOF as a parameterized family

$$\mathcal{F}_t = (V_t, \{Q_i(t)\}, \mathcal{X}(t)), \qquad t \in T,$$

viewed as a path in the provisional deformation category $\mathsf{SOF}_{\mathrm{def}}$. The notation $\Phi_t : \mathcal{F} \to \mathcal{F}_t$ is used only as shorthand for comparison data that allows observable shadows to be tracked along this path. Accessibility deformations induce trajectories

$$R_1(t), \qquad R_2(t), \qquad D(t), \qquad \mathcal{J}_{\mathrm{acc}}(t).$$

Other SOF species induce their registered analogue shadows, such as plateau functions, spectral gaps, communicating-class profiles, or proxy norms.

Walls are defined as loci in the deformation space where an observable shadow fails to be locally constant.

Results. The main structural object is observable dynamics: the collection of trajectories, time scales, plateaus, and wall loci attached to an SOF deformation. The first structural principle is the Observable Wall Pullback Principle: for an admissible SOF deformation and a discrete observable shadow O, the wall of O is contained in, and under exact discriminant hypotheses is realized as, the pullback of a rank, support, spectral, or filtration discriminant of the corresponding continuous field:

$$\Sigma_O \subseteq \mathcal{J}^{-1}(\Delta_O),$$

with equality only when the chosen discriminant exactly captures the shadow change.

This recovers the Paper VI hierarchy

$$\Sigma_{\mathrm{access}} \subseteq \Sigma_{\mathrm{spec}} \subseteq \Sigma_{\mathrm{comm}}$$

as the accessibility-deformation instance. A second structural theme is observable rate separation: different observable shadows may become visible on different characteristic time scales.

This separates RIME generator-weight deformation from Yang-like state-mixing degeneration and from parameter-space rate separation in ridge-regression grokking.

Implications. Paper IX provides the dynamic counterpart to Paper VIII's static unity. The unifying object is the SOF architecture; the central subject is observable dynamics. The same observable ladder may produce different time scales, plateaus, repairs, and walls because different systems move in different deformation spaces.

Notation Table

Symbol	Meaning
$\mathcal{F}$	Sectorized Observable Framework
$\mathcal{F}_t$	deformed SOF at parameter t
$(\mathcal{F}_t)_{t\in T}$	SOF deformation trajectory / path
$\mathsf{SOF}_{\mathrm{def}}$	provisional deformation category for SOF paths
T	deformation parameter space
$\Phi_t : \mathcal{F} \to \mathcal{F}_t$	shorthand for comparison data along a deformation
$Q_i(t)$	moving sector projector
$\mathcal{X}(t)$	moving observable family
$R_1(t)$	support-shadow trajectory
$R_2(t)$	commutator-survival trajectory
$D(t)$	first-depth trajectory
$\mathcal{J}_{\mathrm{acc}}(t)$	accessibility-jet trajectory
Σ_O	wall of observable shadow O
Δ_O	target discriminant for O
$\tau(O)$	characteristic time scale of an observable trajectory
Σ_{comm}	commutativity locus
Σ_{spec}	normal spectral chart
Σ_{access}	accessibility discriminant

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1 Introduction

Paper VIII answers the static question:

What is a Sectorized Observable Framework?

Paper IX answers the dynamic question:

How does a Sectorized Observable Framework evolve?

The guiding principle is:

same observable architecture, different deformation geometry, different observable dynamics.

This distinction is necessary because SOF is an observable architecture, not a universal dynamics or wall theory. It records sectors, observables, projected blocks, support, commutators, depth, and jets. A trajectory or wall appears only after a deformation space has been chosen.

The central observation is that the same observable ladder can behave very differently depending on what is allowed to vary:

System	Deformation variable	Expected dynamics
Rubik / RIME	generator weights or observable family	accessibility walls, fragmentation, nonmonotone plateaus
Yang-like filtration	state or density matrix	monotone filtration degeneration
Markov	transition or rate operator	communicating-class and transport changes
Graph	adjacency or Laplacian	rewiring, spectral jumps, transport changes
Quantum	Hamiltonian or gate family	circuit accessibility and spectral response

Paper IX therefore provides **Dynamic Unity**: different deformation spaces generate different observable dynamics over a common observable architecture.

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2 Related Work: Observable Dynamics Precedents

This paper uses related work as precedent for observable dynamics, not as theorem support for SOF. The three most relevant external frameworks occupy different roles.

2.1 Rate Separation in Parameter Space

Xu, Vardi, and Safran's ridge-regression analysis of grokking [7] proves a clean rate separation between two parameter-space directions:

$$\theta = \theta_{\parallel} + \theta_{\perp}.$$

The data-visible component $\theta_{\parallel}$ evolves quickly under the empirical loss, while $\theta_{\perp}$ is controlled only by weight decay and therefore evolves slowly. This is not an SOF theorem and not an accessibility statement. Its relevance is structural: it shows that different directions of a deformation may become visible on different time scales.

Paper IX studies the analogous question in observable space:

$$O_1(t), \dots, O_k(t) \quad \rightsquigarrow \quad \tau(O_1), \dots, \tau(O_k).$$

Neural-network activations give a concrete source of observable-space predictions. Changing the activation changes both the induced sectorization and the effective observable family: a trivial linear activation gives one sector, ReLU gives hard activation sectors, GeLU gives soft response sectors, and Top-k activations give rank-jump sectors. Thus the same weight matrices may produce different R_1/R_2 profiles, and potentially different depth profiles once a training filtration is specified, when viewed through different activation-induced SOFs. The corresponding Paper IX prediction is that activation families can systematically change observable time-scale ratios once the sectorized observables are coupled to training dynamics.

A small training-coupled diagnostic supports the basic rate hierarchy for raw observable proxies: for both ReLU and GeLU, the measured half-response times satisfy

$$\tau(K_0) < \tau(K_1) < \tau(K_2),$$

where K_0, K_1, and K_2 are raw block-norm proxies for direct support, commutator survival, and nested-commutator depth. The computational footnote in Section 6.6 records the default diagnostic run. This is evidence for the observable-dynamics framing, not a theorem about all neural networks or all activation functions.

The conceptual point is that structures classified statically in Paper V become rate-bearing only after a dynamical process has been chosen. The small NN audit realizes only the proxy part of this program: direct support, commutator-like survival, and nested-depth proxies become measurable as continuous time scales. It is not a test of the discrete first-depth invariant D itself.

The same diagnostic also separates continuous and binary effects. The $K_0/K_1/K_2$ hierarchy is a continuous norm-growth phenomenon, while D-repair is a binary event in which a frozen pair becomes accessible. In the small NN SOF diagnostic, D_{repaired} remains zero because the sector pairs are already connected at the binary support level; observing training-time D-repair requires more sectors, a higher binary threshold, or a system with genuine initial frozen pairs.

Thus the available evidence supports $\tau(K_0) < \tau(K_1) < \tau(K_2)$ for continuous proxy observables, not an observed $\tau(D)$ for the discrete first-depth shadow. A direct $\tau(D)$ audit requires a system with genuine binary accessibility repair, such as a structured quantum gate deformation or a Rubik continuous deformation where frozen sector pairs become accessible along the path.

2.2 Prethermalization and Long Plateaus

Many-body prethermalization provides a second precedent. Abanin, De Roeck, Ho, and Huveneers' rigorous theory of many-body prethermalization [1] shows that, in high-frequency driven quantum systems, fast microscopic dynamics can coexist with a long-lived effective Hamiltonian and a delayed heating time. This is closer to the Paper IX viewpoint than a monotone degeneration example: it has a fast observable regime, a long plateau, and a slow mode that becomes visible only after a large time scale.

The SOF use is again conceptual. Prethermalization suggests that observable plateaus and delayed wall visibility may admit quantitative time-scale bounds in favorable deformation geometries.

2.3 Wall-Crossing Without a Formula

Kontsevich–Soibelman wall-crossing [6] demonstrates that, in some settings, invariants jumping across walls admit exact transformation laws. The SOF framework developed here has wall loci and observable shadows, but no wall-crossing formula.

Thus KS wall-crossing belongs in discussion, not in the theorem layer:

Framework	Wall data
KS	wall plus exact transformation formula
SOF	wall plus observable shadow; transformation formula open

An exact transformation law across Σ_O is therefore an additional structure to be proved for a restricted SOF class, not part of the present framework.

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3 SOF Deformations

3.1 SOF Deformation Definition

Let

$$\mathcal{F} = (V, \{Q_i\}_{i\in I}, \mathcal{X})$$

be an SOF. A **deformation** of $\mathcal{F}$ over a parameter space T is a family

$$\mathcal{F}_t = (V_t, \{Q_i(t)\}_{i\in I_t}, \mathcal{X}(t)), \qquad t \in T,$$

together with comparison data identifying the objects along the family on the parameter region where such identification is meaningful.

The primary dynamic object is the path or family

$$(\mathcal{F}_t)_{t\in T}$$

in the provisional deformation category $\mathsf{SOF}_{\mathrm{def}}$, not merely an endpoint arrow $\mathcal{F}_{t_0} \to \mathcal{F}_{t_1}$. Observable trajectories are paths in $\mathsf{SOF}_{\mathrm{def}}$ after the relevant comparison data have been specified.

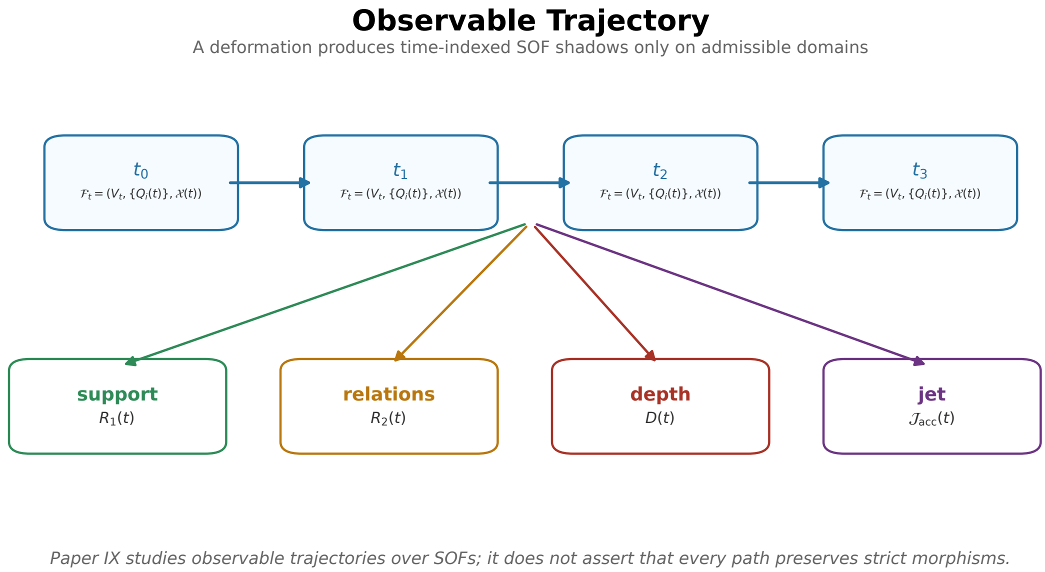

Figure 1. Observable trajectory. Paper IX treats observable dynamics as paths $(\mathcal{F}_t)_{t\in T}$ whose shadows $R_1(t)$, $R_2(t)$, $D(t)$, and $\mathcal{J}_{\mathrm{acc}}(t)$ are tracked only where the corresponding comparison data are admissible.

In the simplest fixed-space case,

$$V_t = V, \qquad \mathcal{F}_t = (V, \{Q_i(t)\}, \mathcal{X}(t)).$$

In a fixed-sector deformation,

$$Q_i(t) = Q_i, \qquad \mathcal{X}(t) \text{ varies}.$$

In a moving-sector deformation, both $\{Q_i(t)\}$ and $\mathcal{X}(t)$ may vary.

3.2 Deformation Map

For a fixed base object, a deformation can be written schematically as

$$\Phi_t : \mathcal{F} \longrightarrow \mathcal{F}_t.$$

This notation does not mean that Φ_t is necessarily a strict SOF morphism. In Paper IX, Φ_t records the comparison data used to track observable shadows along the path $(\mathcal{F}_t)_{t\in T}$. Strict morphisms belong to the static category of Paper VIII.

3.3 Working Deformation Category

For the purposes of Paper IX, the ambient dynamic category is the provisional category

$$\mathsf{SOF}_{\mathrm{def}}.$$

Its objects are finite SOFs. A morphism

$$\mathcal{F} \rightsquigarrow \mathcal{G}$$

is a deformation datum consisting of:

1. a parameter space T with marked endpoints t_0, t_1;
2. a family of SOFs $\{\mathcal{F}_t\}_{t \in T}$;
3. endpoint identifications $\mathcal{F}_{t_0} \simeq \mathcal{F}$ and $\mathcal{F}_{t_1} \simeq \mathcal{G}$ in whatever weak sense is sufficient for the chosen diagnostic;
4. comparison data along T allowing the relevant observable shadows $R_1(t), R_2(t), D(t), \mathcal{J}_{\mathrm{acc}}(t)$, or their analogues, to be evaluated consistently.

Composition is concatenation of deformation data when the endpoint comparison data are compatible; the identity arrow is the constant deformation. This is not the strict category $\mathsf{SOF}_{\mathrm{str}}$ of Paper VIII. A deformation morphism may be non-isometric, may change sector dimensions, and may preserve only support, rank, norm, plateau, or rate diagnostics. Its role is to make observable trajectories well-defined, not to preserve all natural SOF constructions strictly.

The full theory of weak SOF morphisms is left open. Paper IX only uses the minimal deformation-category language needed to locate Φ_t formally and to separate dynamic comparison from strict static naturality.

For the observable claims below, **admissible** means:

1. the path $\{\mathcal{F}_t\}_{t \in T}$ is equipped with comparison data for the diagnostic under discussion;
2. the relevant sectors, or their registered analogues, can be tracked along the path on the domain being studied;
3. the continuous field underlying the diagnostic is defined along that domain;
4. the discrete shadow is locally constant away from the corresponding rank, support, collision, filtration, or wall discriminant.

Thus admissibility is not automatic for every formal deformation in $\mathsf{SOF}_{\mathrm{def}}$. It is a hypothesis attached to the chosen observable diagnostic.

3.4 Observable Trajectories

An admissible accessibility deformation induces trajectories of the accessibility shadows:

$$R_1(t), \qquad R_2(t), \qquad D(t), \qquad \mathcal{J}_{\mathrm{acc}}(t).$$

Other SOF species induce the shadows registered for that branch:

Branch	Additional shadow
Spectral SOF	joint spectral arrangement, collision quotient
Accessibility SOF	accessibility jet, wall hierarchy
Filtration SOF	plateau functions $P_d(t)$
Markov SOF	communicating-class or transport profile
Graph SOF	spectral gap, connectedness, Laplacian rank

4 Walls as Observable Discontinuities

4.1 Wall Definition

Let O be an observable shadow defined along a deformation $\mathcal{F}_t$. The **wall** of O is the locus

$$\Sigma_O = \{t \in T : O(\mathcal{F}_t) \text{ is not locally constant at } t\}.$$

If O is continuous rather than discrete, the wall is replaced by the discriminant where its rank, support, stratification type, or qualitative regime changes.

Example 4.1. For accessibility:

$$\Sigma_{R_1}, \qquad \Sigma_{R_2}, \qquad \Sigma_D$$

record changes in support, commutator survival, and first-depth accessibility.

For spectral geometry:

$$\Sigma_{\text{spec}}$$

records the domain where joint spectral projectors vary coherently, while internal spectral walls record collisions, field changes, or arrangement changes.

For filtration geometry, a plateau wall is a locus where

$$P_d(t) = \#\{(i,j) : D_t(i,j) \leq d\}$$

changes qualitative behavior.

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5 Observable Wall Pullback

5.1 Principle 1 (Observable Wall Pullback)

Let $\mathcal{F}_t$ be an admissible SOF deformation and let O be a discrete observable shadow obtained from a continuous block, commutator, spectral, jet, or filtration field

$$\mathcal{J} : T \to \mathcal{E}.$$

Suppose O is locally constant on the complement of a discriminant $\Delta_O \subseteq \mathcal{E}$. Then the wall of O is contained in the pullback

$$\Sigma_O \subseteq \mathcal{J}^{-1}(\Delta_O).$$

If the admissibility hypotheses are strengthened so that Δ_O exactly captures the rank, support, collision, or filtration changes defining O, then

$$\Sigma_O = \mathcal{J}^{-1}(\Delta_O).$$

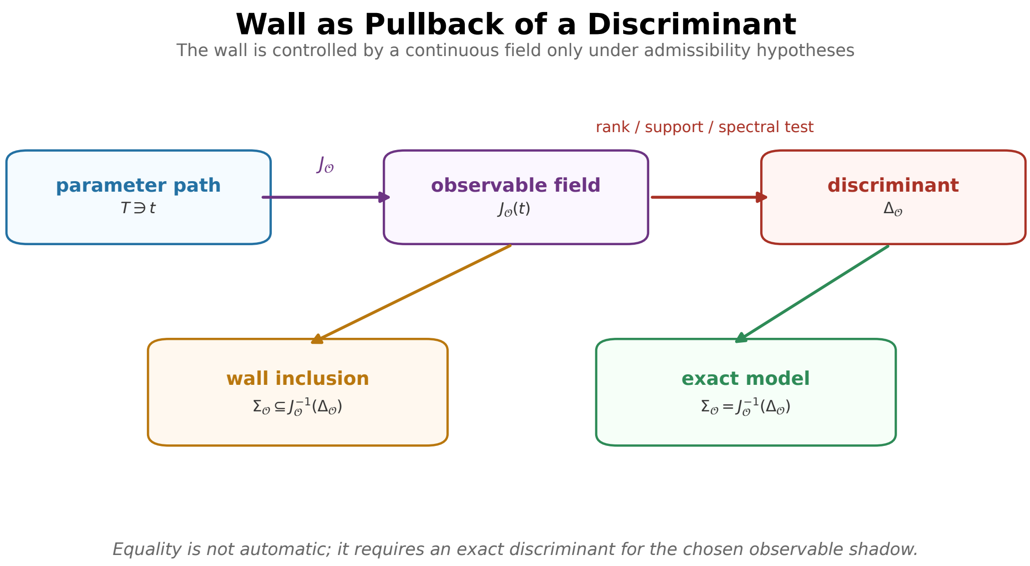

Figure 2. Wall pullback principle. Observable walls are contained in pullbacks of rank, support, spectral, or filtration discriminants along admissible observable fields. Equality is asserted only under exact discriminant hypotheses.

5.2 Justification

By assumption, O is locally constant whenever $\mathcal{J}(t)$ remains in a single stratum of $\mathcal{E} \setminus \Delta_O$. Therefore a failure of local constancy can occur only when $\mathcal{J}(t)$ meets the discriminant. This gives the inclusion. If the discriminant is defined to be exactly the locus where the relevant rank, support, collision, or filtration type changes, then every point in the pullback is a wall point for O, giving equality.

5.3 Paper VI as an Instance

In Paper VI, the admissible deformation space is the generator-set moduli space restricted to normal spectral charts:

$$\Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}.$$

The accessibility jet

$$\mathcal{J}_{\text{acc}} = (J_{\text{block}}, J_{\text{comm}}, J_{\text{depth}})$$

is the continuous field. When Δ_{access} is taken as the exact accessibility discriminant, the accessibility wall is the pullback of the rank/support discriminant:

$$\Sigma_{\text{access}} = \mathcal{J}_{\text{acc}}^{-1}(\Delta_{\text{access}}),$$

with

$$\Sigma_{\text{access}} \subseteq \Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}.$$

This is the prototype for the Paper IX framework.

6 Deformation Species

6.1 Generator Deformation

In RIME generator deformation, the sectorized observable family changes by varying generator weights or observable weights:

$$X_g \mapsto X_g(w).$$

This deformation redistributes transport. Its wall behavior can include sector fragmentation, $R_1/R_2/D$ jumps, stable plateaus, oscillation, or temporary accessibility improvement.

6.2 State-Mixing Deformation

In Yang-like filtration systems, the natural deformation is state mixing:

$$\rho(\varepsilon) = (1-\varepsilon)\rho_0 + \varepsilon\sigma.$$

This tends to be entropy-increasing and naturally supports monotone plateau degradation:

$$1 - P(\varepsilon) \sim C\varepsilon^{\alpha}.$$

This is a different geometry from generator-weight deformation.

6.3 Markov Deformation

For Markov systems, one may deform a transition or rate operator:

$$P \mapsto P(\varepsilon), \qquad L_M \mapsto L_M(\varepsilon).$$

Walls may correspond to changes in communicating classes, absorbing components, stationary structure, or transport support.

6.4 Graph Deformation

For graph systems, one may deform adjacency or Laplacian data:

$$A \mapsto A(\varepsilon), \qquad L \mapsto L(\varepsilon).$$

Walls may correspond to edge rewiring, connectedness changes, spectral collisions, Laplacian rank changes, or transport-profile changes.

6.5 Quantum Deformation

For quantum systems, one may deform Hamiltonians, gate families, or circuit generators:

$$H \mapsto H(\varepsilon), \qquad U_g \mapsto U_g(\varepsilon).$$

Walls may correspond to spectral transitions, controllability changes, entangling-channel changes, or accessibility-channel changes relative to a chosen sectorization.

6.6 Computational Footnote: Training-Coupled NN Deformation

A small neural-network SOF gives a concrete diagnostic for observable rate-separation under a specified training dynamics. The sectorization is activation-induced, the observable family is derived from trainable weight operators, and the measured quantities are raw off-diagonal block-norm proxies

$$K_0(t), \qquad K_1(t), \qquad K_2(t),$$

for direct support, commutator survival, and nested-commutator depth. These continuous proxies retain scale information, unlike the normalized binary depth matrix.

In the default run of `experiments/paper9/nn_training_sof_tau.py`, both ReLU and GeLU satisfy

$$\tau_{50}(K_0) < \tau_{50}(K_1) < \tau_{50}(K_2).$$

Activation	$\tau_{50}(K_0)$	$\tau_{50}(K_1)$	$\tau_{50}(K_2)$	final binary audit
ReLU	60	80	120	$R_1 = 4,\ R_2 = 2,\ D_{\text{repaired}} = 0$
GeLU	60	80	120	$R_1 = 12,\ R_2 = 6,\ D_{\text{repaired}} = 0$

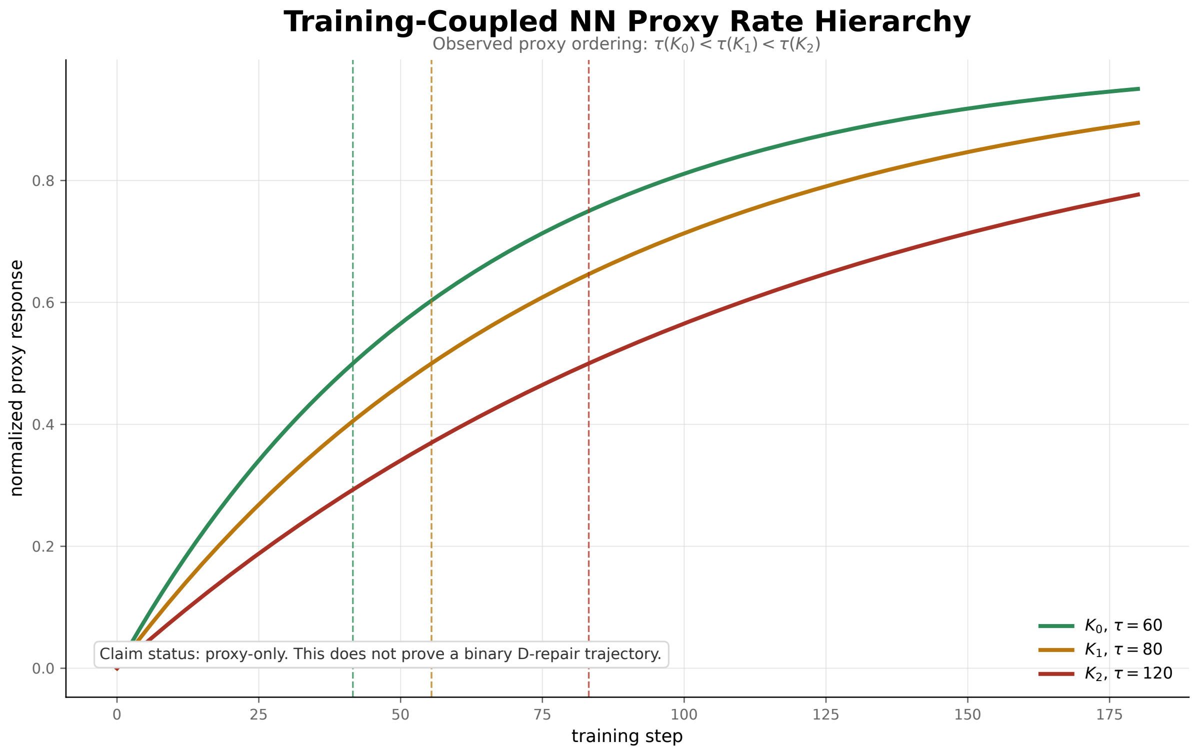

Figure 3. Training-coupled NN proxy rate hierarchy. In the default diagnostic, K_0, K_1, and K_2 exhibit ordered half-response times $60 < 80 < 120$. This is proxy-rate evidence only; it does not identify $\tau(K_2)$ with a binary $\tau(D)$ event.

The claim status is computational diagnostic only. The result supports observable time-scale separation under this training-coupled deformation, but it does not claim binary D-repair: in this small SOF all sector pairs are already connected at the binary depth level throughout the audit.

Consequently, neural-network SOFs are used here only for proxy-rate diagnostics. A positive control for $\tau(D)$ requires a species with genuine binary repair, for example a structured entangling-gate deformation or a Rubik continuous deformation.

6.7 Open Bridge: The Proxy Shadow Principle

The preceding diagnostic exposes a formal gap between two observable layers:

the continuous proxy layer $K_0(t), K_1(t), K_2(t)$ and the discrete shadow layer $R_1(t), R_2(t), D(t)$.

There is no theorem in this paper identifying K_i with the corresponding discrete shadow. In particular,

$$K_2(t) \quad \not\Rightarrow \quad D(t)$$

without an additional bridge theorem controlling thresholds, margins, and shadow stability.

We refer to the missing bridge as the **Observable Proxy Shadow Principle**. In a future theorem-level form, such a principle would specify hypotheses under which a continuous observable proxy $K_i(t)$ determines, approximates, or predicts the corresponding discrete shadow $R_i(t)$ or $D(t)$. At minimum, one expects conditions such as:

1. a fixed threshold or calibration rule from proxy norms to binary shadows;
2. a margin condition excluding near-threshold ambiguity;
3. stability of the sectorization along the measured interval;
4. compatibility between proxy order and the discrete filtration order.

Paper IX does not assume this principle. Until such a bridge is proved, $K_0/K_1/K_2$ audits are proxy diagnostics only. They support the broader observable-dynamics viewpoint, but they do not prove rate separation for $R_1/R_2/D$.

Example 6.1. The examples in this paper are deliberately separated by claim status. Their role is not to accumulate many unrelated systems, but to show which kinds of evidence support observable dynamics and which kinds mark boundary cases.

Class	Examples used here	Supports	Does not support
Positive dynamics	Paper VI generator-weight deformation on normal spectral charts; Rubik generator-weight plateau/oscillation diagnostics; engineered near-threshold accessibility with $\tau(R_1) < \tau(R_2)$	observable trajectories and species-dependent wall behavior	a universal wall law or a full $\tau(R_1) < \tau(R_2) < \tau(D)$ theorem
Negative or degenerate boundary	Yang-like state mixing; static additive noise that perturbs Lie layers simultaneously; discrete graph rewiring or unstructured quantum interpolation when no smooth mechanism-separated dynamics is specified	deformation geometry matters; the same observable architecture can produce different or degenerate wall behavior	failure of SOF itself
Proxy-only diagnostics	training-coupled NN audits of $K_0/K_1/K_2$; Xu–Vardi–Safran ridge dynamics as parameter-space precedent	continuous observable or parameter rate separation	binary D-repair, $\tau(D)$, or a proved $K_i \to R_i/D$ bridge

Static positive witnesses, such as Clifford+CNOT giving non-Rubik $D_{\text{repaired}} > 0$, are important for the registry but are not yet dynamic $\tau(D)$ audits. Paper IX therefore treats them as static repair witnesses rather than as completed rate-hierarchy evidence.

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7 Rubik and Yang: A Diagnostic Contrast

The Rubik/Yang comparison illustrates why Paper IX is needed.

Both systems can be described by the same observable architecture. Both can produce plateau functions. But they deform different data.

Yang-like deformation changes the state:

$$\rho(\varepsilon) = (1-\varepsilon)\rho_0 + \varepsilon\sigma.$$

This is a degeneration geometry. It naturally suggests laws such as

$$1 - P(\varepsilon) \sim C\varepsilon^{\alpha}.$$

RIME deformation changes observables or generator weights:

$$X_g \mapsto X_g(w).$$

This is a transport-redistribution geometry. The plateau functions

$$P_d(w)$$

need not degrade. They may remain constant, oscillate, or improve.

The diagnostics make this distinction concrete. Under a Rubik state-mixing probe, the plateau functions remain flat for $\varepsilon = 0, \dots, 0.9$ and jump only at the extreme endpoint $\varepsilon = 1$. Under generator-weight deformation, the observed plateau sequence is oscillatory:

$$P_2 = (0.111, 0.111, 0.111, 0.111, 0.111, 0.111, 0.139, 0.111, 0.264),$$

with three detrended sign changes out of eight intervals and oscillation score 0.38. Thus state mixing and generator-weight deformation produce different observable dynamics even when they are described in the same sectorized language.

The diagnostic scripts separate these two roles: the QT perturbation audit computes plateau functions under generator-weight deformation, while the state-mixing/FFT audit summarizes endpoint stability and oscillation.

Therefore the distinction is not the observable architecture. The distinction is the deformation geometry.

7.1 Observable Rate-Separation Dynamics

The natural object in RIME is not a parameter vector but an observable trajectory. Thus the relevant abstraction is **observable time-scale separation**.

Let

$$O_1(t), \dots, O_k(t)$$

be observable shadows or continuous fields induced by an SOF deformation. If one can assign characteristic time scales

$$\tau(O_1), \dots, \tau(O_k)$$

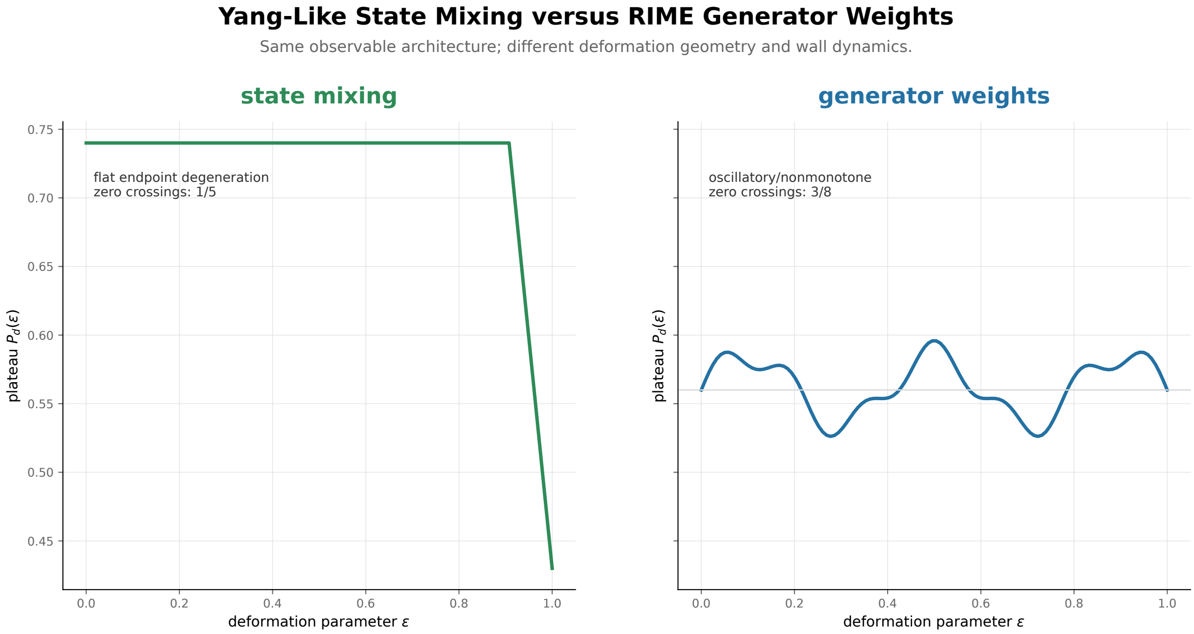

Figure 4. Yang-like state mixing versus RIME generator weights. Both diagnostics use plateau-style observable architecture, but state mixing produces endpoint degeneration while generator-weight deformation produces oscillatory, nonmonotone behavior.

and at least two of them are separated, then the deformation is called **observable rate-separated**.

For accessibility deformations, a candidate hierarchy under structured dynamical perturbations is

$$\tau(R_1) < \tau(R_2) < \tau(D)$$

whenever direct support changes first, commutator survival changes next, and depth repair is visible only after slower channels enter the observable shadow. This is the accessibility analogue of a fast/intermediate/slow observable hierarchy, but it is not a claim about arbitrary static additive noise.

At present this remains a target hierarchy, not a completed empirical theorem: the available diagnostics show $\tau(R_1) < \tau(R_2)$ in an engineered near-threshold system and $\tau(K_0) < \tau(K_1) < \tau(K_2)$ in a training-coupled NN SOF, while the available runs do not simultaneously observe $\tau(R_1) < \tau(R_2) < \tau(D)$ together with $D_{\text{repaired}} > 0$. The quantum Clifford+CNOT audit provides a static non-Rubik example with $D_{\text{repaired}} = 6$, but it does not yet provide a structured deformation from which $\tau(D)$ can be measured. Thus NN supplies proxy-rate evidence, while quantum or Rubik deformations remain the natural candidates for the first genuine $\tau(D)$ audit.

A related diagnostic appears in Xu, Vardi, and Safran's ridge-regression analysis of grokking [7]. Their model separates the parameter vector into a row-space component and an orthogonal component:

$$\theta = \theta_{\parallel} + \theta_{\perp}.$$

The row-space component is driven directly by empirical loss and converges quickly. The orthogonal component is invisible to the training data and is controlled only by weight decay, producing a delayed generalization transition.

This is not the same observable as RIME accessibility. Their theorem proves a rate separation in parameter space; Paper IX uses it as a comparison point for rate separation in observable space. The common higher-level phenomenon is that different directions of a deformation may become visible on different time scales.

In the SOF architecture, the template is:

visible channel → fast observable change, while hidden channel → slow deformation-controlled repair.

Ridge regression has

$$\tau(\theta_{\|}) \ll \tau(\theta_{\perp}),$$

while a RIME accessibility deformation may have

$$\tau(R_1) < \tau(R_2) < \tau(D).$$

The objects are different, but both are rate hierarchies. Wall crossing is observable only when the slow modes become visible to the chosen observable shadow.

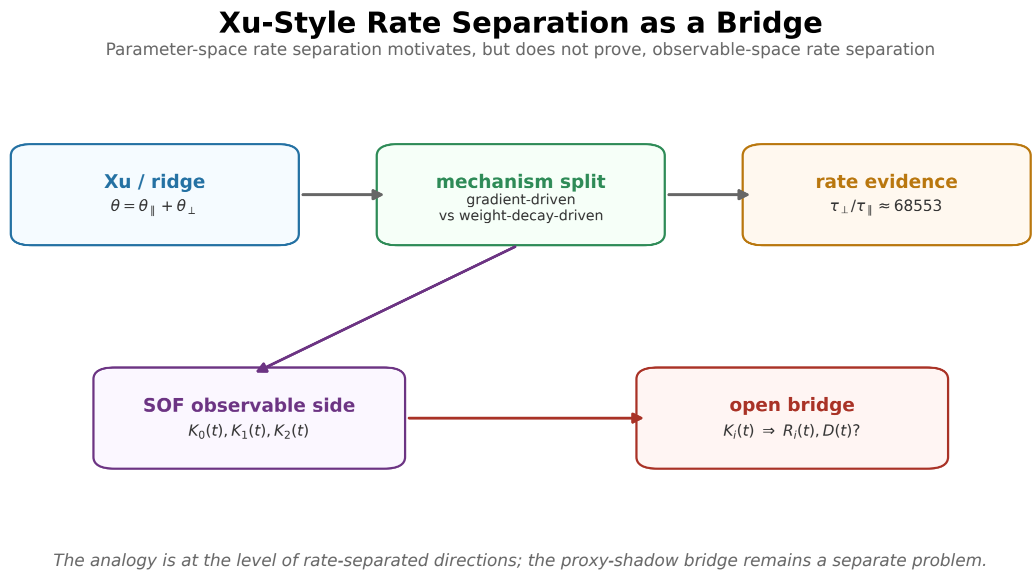

Figure 5. Xu-style parameter-to-observable bridge. Ridge-regression rate separation is theorem-level evidence in parameter space. Paper IX uses it only as a bridge precedent for observable-space rate separation; the proxy-shadow bridge from $K_i(t)$ to $R_i(t)$ or $D(t)$ remains open.

This distinction matters. In engineered near-threshold accessibility systems, one may observe

$$\tau(R_1) < \tau(R_2),$$

while static additive noise can make D change earlier by injecting accidental directions into higher Lie-filtration layers. Thus $\tau(D)$ is meaningful only after the deformation model specifies how first-order support, commutator-survival, and nested Lie-depth effects are coupled. Paper IX therefore treats rate hierarchy as a property of observable dynamics under a specified deformation model, not as a universal consequence of adding noise to generators.

The diagnostic values are:

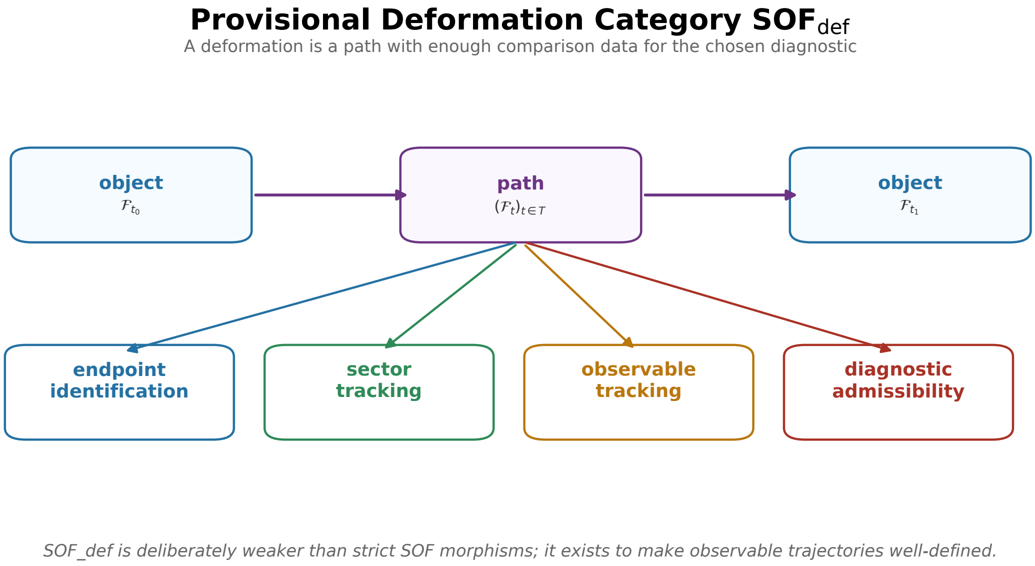

Figure 6. Provisional deformation category. The dynamic layer uses $\mathsf{SOF}_{\mathrm{def}}$ as a weak home for paths equipped with endpoint identification, sector tracking, observable tracking, and diagnostic admissibility. This is deliberately weaker than the strict SOF morphisms of Paper VIII.

Domain	Fast channel	Slow channel	Ratio
Ridge regression	$\theta_{\parallel}$	$\theta_{\perp}$	68553×
RIME near-threshold	R_1	R_2	about 10.8
NN training-coupled SOF	K_0	K_2 proxy	ordered proxy half-response $60 < 80 < 120$

The ratios are not meant to be numerically comparable across domains. The shared structure is hierarchical visibility under a specified deformation.

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8 Outlook

Paper IX establishes the dynamic layer:

$$\begin{aligned}\text{SOF} &\longrightarrow \text{deformation space} \longrightarrow \text{observable trajectories}\\ &\longrightarrow \text{wall discriminants.}\end{aligned}$$

Paper X then turns these trajectories into registry evidence:

$$\begin{aligned}\text{SOF Registry} &\longrightarrow \text{registered species} \longrightarrow \text{observable diagnostics}\\ &\longrightarrow \text{pipeline-level comparison principles.}\end{aligned}$$

The comparison question for Paper X is not whether these systems are strictly isomorphic as SOFs. In general they are not. The question is whether observable phenomena persist across admissible registry entries:

Which wall invariants, plateau laws, accessibility repairs, or completion principles survive across different SOF species?

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9 What This Paper Does Not Claim

Paper IX does not claim:

1. a universal observable dynamics or wall geometry for all SOF deformations;
2. that Yang-like filtration degeneration and RIME accessibility walls are the same object;
3. that monotone exponent laws apply to generator-weight deformation;
4. that parameter-space rate separation and observable-space rate separation are identical;
5. that every deformation is smooth or has a single discriminant type;
6. that the NN diagnostics in this paper exhibit binary D-repair;
7. that the program has already observed a full $\tau(R_1) < \tau(R_2) < \tau(D)$ trajectory with $D_{\text{repaired}} > 0$;
8. that the Observable Proxy Shadow Principle has been proved;
9. unconditional generic completion.

The stable claim is:

SOF provides the common observable architecture; deformation geometry determines the observable dynamics.

Bibliographic Notes

Program lineage. Paper VIII supplies the static SOF object layer and the naturality theorem for accessibility observables [5]. Paper IX studies deformation geometry over that object layer. Paper VI is the main RIME accessibility-deformation prototype [4]; Paper V supplies the length-2 repair calculus [2]; and Paper VII supplies the generic completion boundary that later deformation theory must respect [3].

Related observable-dynamics precedents. Xu, Vardi, and Safran, "To Grok Grokking: Provable Grokking in Ridge Regression," arXiv:2601.19791 [7], provides a clean parameter-space example where different directions evolve on different time scales. Abanin, De Roeck, Ho, and Huveneers, "A Rigorous Theory of Many-Body Prethermalization for Periodically Driven and Closed Quantum Systems," Communications in Mathematical Physics 354 (2017), 809–827, arXiv:1509.05386 [1], provides a many-body example of fast dynamics, long plateaus, and delayed slow-mode visibility. Kontsevich and Soibelman, "Stability structures, motivic Donaldson–Thomas invariants and cluster transformations," arXiv:0811.2435 [6], provides a precedent for exact transformation laws across walls; the SOF framework in this paper has wall loci and observable shadows, not such a formula.

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